Project Proposal

PROJECT NAME	Internal Assistant for Enterprise Knowledge (Slack App Bot)		
PROJECT LOCATION	Swinburne University of Technology		
EST. START DATE	Sep 29th, 2026	[redacted]	Dec 6th, 2026
PROJECT LEADER	Mr. Truong Ngoc Gia Hieu	**UNIVERSITY**	Swinburne University of Technology
CONTACT NAME	Mr. Truong Ngoc Gia Hieu	**ADDRESS**	1A/12 Bach Dang Street, Tan Son Hoa Ward, Ho Chi Minh City
PHONE	(+84)947979438		
EMAIL	giahieutruongngoc@gmail.com		

SUMMARY	The project addresses the challenge of fragmented internal knowledge at Vietnam International Bank (VIB). In particular, vital knowledge distributed across multiple Confluence pages and internal documentation sites creates time consuming information retrieval processes for bank personnel. Consequently, this challenge necessitates a centralized knowledge assistant integrated directly into Slack utilizing Retrieval Augmented Generation technology. The proposed solution enables employees to query internal documentation using natural language interfaces. Therefore, the primary scope of this project encompasses building a unified knowledge retrieval platform featuring automated document ingestion, vector indexing, and channel level access control mechanisms.
DESIRED OUTCOME	- The project aims to deliver a functional Slack integrated knowledge assistant prototype in order to enable VIB employees to access organizational knowledge stored in Confluence through natural language queries. - Specifically, the assistant will retrieve information from employee accessible Confluence content, including company information, internal policies, employee benefits, banking products, and technical architecture documentation. Furthermore, users are expected to receive relevant, source based answers directly within Slack. Consequently, this reduces the need to manually search through multiple Confluence pages. - In addition, project success will be evaluated based on answer accuracy, retrieval relevance, response time, and compliance with defined Slack channel restrictions. Ultimately, the expected outcome is a more accessible and efficient way for VIB employees to find and use internal knowledge in their daily work.

ACTION TO COMPLETION	- Multi Source Ingestion and Indexing: First and foremost, develop an automated pipeline to collect, chunk, and index internal documentation from multiple sources into a vector database. - RAG Engine Implementation: Subsequently, set up the Retrieval Augmented Generation core to process natural language queries and retrieve precise answers. - Slack App Bot Integration and Whitelist Configuration: Next, build the official conversational interface as a Slack app bot and configure channel level whitelist restrictions to limit usage to designated channels within the private corporate workspace. - System Testing and Validation: Finally, perform end to end testing of document retrieval accuracy, response latency, and channel security boundaries.
BENEFITS OF PROJECT	The deployment of the proposed knowledge assistant delivers substantial strategic, operational, and technical advantages for Vietnam International Bank (VIB), structured across three primary dimensions: • Elimination of Operational Knowledge Retrieval Bottlenecks: First and foremost, the system eliminates time wasted searching across fragmented documentation by providing instant, source backed answers directly within the primary team communication tool via a dedicated Slack app bot. Consequently, this minimizes context switching overhead for bank personnel, reduces mean time to information resolution, and significantly enhances daily workforce productivity across operational departments. • Extensible Architecture and Multi Source Ingestion Value: Furthermore, the platform delivers immediate operational value through multi source document support, seamlessly aggregating knowledge from Confluence spaces, technical repositories, and internal policy guides. Simultaneously, it establishes a robust, highly extensible technical foundation, allowing future expansion into additional corporate knowledge repositories, enterprise databases, and automated workflow integrations without requiring core architectural redesigns. • Enterprise Grade Privacy, Identity, and Access Governance: In addition, the architecture ensures high enterprise data privacy and strict information security compliance by operating entirely within a private corporate workspace. By leveraging enterprise Slack Single Sign On (SSO) for identity authentication and enforcing simple, robust channel whitelist restrictions, the system guarantees that sensitive internal documentation remains accessible strictly to authorized user groups and designated operational channels.
PROJECTED SCHEDULE	- Week 5 (Oct 12 – Oct 18): Documentation Finalization & Infrastructure Research • Milestone: Project Governance Established & Database Alternatives Evaluated • Finalize and sign-off on core project documentation, including the project charter and project proposal. • Initialize the official project GitHub repository, establish branching strategies, and configure team access permissions. • Research and evaluate database alternatives to avoid complex Linux and Ubuntu server configurations, focusing on lightweight options that can run smoothly locally. - Week 6 (Oct 19 – Oct 25): Architecture Design Finalization & Environment Setup • Milestone: Complete System Architecture Designed & Local Environment Initialized

- Finalize the comprehensive system architecture design, mapping out the interactions between document ingestion, vector storage, the retrieval-augmented generation engine, and the Slack app bot interface.
- Settle definitively on the local database deployment architecture and containerization approach to streamline team onboarding.
- Initialize the local development environment for concurrent setup across frontend, backend, database, and language model components.

- Week 7 (Oct 26 – Nov 01): Core Ingestion Pipeline & Security Foundation
  - Milestone: Document Ingestion Scripting & Channel Whitelist Implemented
  - Concentrate engineering manpower on building the document ingestion pipeline to parse public and low-level corporate documentation based on the finalized architecture.
  - Implement the foundational channel-level security whitelist mechanism early in development to restrict bot interactions strictly to designated testing channels.
  - Establish the baseline modular folder structure in the GitHub repository to ensure clean team synchronization and prevent merge conflicts.
- Week 8 (Nov 02 – Nov 08): RAG Engine & LLM Integration
  - Milestone: Retrieval-Augmented Generation Backend Functional
  - Focus team manpower on connecting the vector database retriever to the language model backend service using local developer configurations.
  - Build prompt engineering templates with strict system instructions to guarantee that response generation includes precise source citations and references as requested by the client.
  - Test baseline retrieval accuracy against sample public enterprise queries.
- Week 9 (Nov 09 – Nov 15): Slack App Bot Interface & Real-Time Messaging
  - Milestone: Interactive Slack App Bot Prototype of Concept (PoC) Working
  - Shift development focuses toward the frontend communication layer by implementing the Slack App Bot interface utilizing the Slack Bolt API and Socket Mode.
  - Configure Socket Mode to enable real-time local messaging without requiring complex public webhooks or live server deployments.
  - Link Slack event handlers directly to the RAG backend engine to enable live conversational Q&A with integrated source citations.
- Week 10 (Nov 16 – Nov 22): Performance Benchmarking & Architecture Future-Proofing
  - Milestone: Response Latency Benchmarks Met & Scalability Framework Ready
  - Execute response latency benchmarks and performance testing against target speed thresholds to ensure the system handles queries efficiently.
  - Refine the data ingestion and database layer design to ensure the architecture remains modular and easily expandable for future integration with alternative enterprise applications.
  - Conduct thorough testing of edge cases, unanswerable queries, and security boundary restrictions.
- Week 11 (Nov 23 – Nov 29): End-to-End System Testing & Refinement
  - Milestone: Integrated System Validation Complete
  - Run comprehensive end-to-end testing across all system layers: document retrieval accuracy, citation correctness, response speed benchmarks, and channel access boundaries.
  - Conduct team-wide code reviews, resolve remaining bugs, and polish user interaction flows within Slack.

	- Week 12 (Nov 30 – Dec 06): Final Polish, Demonstration & Project Handover • Milestone: Deployment-Ready Prototype of Concept Handover & Future Roadmap Finalized • Prepare final demonstration scripts, testing reports, and technical documentation packages for the client and academic mentors.Moreover, Package the working prototype of concept for client review and outline the future scaling roadmap for subsequent project phases.
PROJECTED BUDGET	The financial assessment for the system is structured into two operational phases to align with the engineering roadmap and enterprise cost efficiency requirements: • Phase 1 Proof of Concept and Academic Development Scope: During the current project phase, direct software licensing costs are minimized by adhering to the client strategy of deploying open source frameworks, local compute environments, and existing developer tier APIs. The core stack leverages locally hosted open source models and open source vector databases on student development hardware, resulting in zero additional direct capital expenditure for cloud compute while preserving full functionality for technical verification. • Phase 2 Production Deployment and Enterprise Scaling Scope: Transitioning the verified prototype to a full scale enterprise production environment in subsequent deployment phases will require dedicated operational budgeting. Anticipated future expenses include cloud hosting infrastructure, managed vector database subscriptions, commercial large language model inference tokens, and enterprise level API quotas to support high volume corporate usage across Vietnam International Bank.
PROJECTED TEAM AND RESOURCE REQUIREMENTS	The execution of the proposed project relies on structured human resources, technical development tools, software infrastructure, and academic literature whose specific requirements are detailed as follows: • Human Resources and Team Structure: A dedicated team of computer science students responsible for end to end system delivery. Additionally, the workload is partitioned into system architecture design, Retrieval Augmented Generation pipeline engineering, vector database optimization, Slack bot API integration, and academic documentation. • Development Environment and Core Stack: The software stack utilizes Python as the primary programming language, supported by vector database frameworks such as Qdrant or Milvus for high dimensional embedding storage, orchestration frameworks like LangChain or LlamaIndex, and foundational machine learning libraries for text processing. • API Ingestion and Integration Infrastructure: Official integration tools comprising the Confluence REST API for automated document extraction, Slack Bolt API SDK for conversational channel interaction, and enterprise grade developer tools within private corporate workspaces. • Computational Hardware and Cloud Infrastructure: Local development environments backed by high memory compute instances and secure API access keys for language model inference, ensuring rapid prototyping and scalable pipeline testing. • Academic Literature and Reference Base: A curated repository of peer reviewed IEEE literature, empirical technical reports, and industry publications focusing on Retrieval Augmented Generation architectures,

	channel level whitelist access control models, and enterprise knowledge graph synchronization.

Project Manager's Signature Date: Oct 14th. 2026 	**Client's Signature** Date: Oct 14th, 2026 